



FIT Lecture # 9

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14.3.4 The Internet

The Internet is a global area network that allows computers connected over the network to share resources and information using different protocols. It is basically a network of networks across the globe. Users at different locations can very easily communicate with each other via the Internet. They can access data and information from other computers if they are permitted to do so. The Internet basically uses a set of protocols such as Transmission Control Protocol/ Internet Protocol (TCP/IP) for transferring the data over the network. The following are the different types of services provided by the Internet to the users:

- **E-mail** E-mail is the widely used service of the Internet through which users can send and receive messages in an efficient way. It allows an individual to send the message even if the other individual is offline. The e-mail messages are sent to and from a unique e-mail address. The series of the characters usually in the form username@hostname for sending the messages to a person is known as the e-mail address.
- **Online chat** The online chat generally allows one-to-one communication between the individuals over the Internet. Users can communicate directly by sending and receiving messages to their fellow friends easily. Online chat is possible only when the sender and receiver both are online.
- **Online shopping** The Internet has become popular by providing shopping of different products online. Now, it is very easy and simple to buy the products from the other parts of the country or from anywhere across the world. A user can purchase an item with the use of personal computer, Internet and special softwares such as shopping cart and e-cash. The shopping cart is used for selecting the items online whereas e-cash is used for making the online payment of the items purchased.
- **Usenet** Usenet is the Internet based newsgroup that allows individuals to share their views regarding a specific topic or subject over the network. People accessing the usenet can receive these views through private or public e-mail.

There are various advantages of using the Internet:

- **Information** The Internet allows users to access large amount of information efficiently. The information could be related to any topic, such as law, marketing, finance, science, technology etc.
- **Availability** The Internet is available continuously to the users all the time without any delay. Users can access the information from the Internet at any time.
- **Cost** The Internet provides different facilities to the users at a low cost. Users can access any website over the Internet absolutely free. The cost of sending messages through e-mail is also cheaper as compared to postal messages.

The following are some of the disadvantages of using the Internet:

- **Hacking** The process of illegally accessing the personal information stored over the Internet is called *hacking*. The person who is involved in this activity is called a *hacker*.
- **Virus** The software program that itself gets activated in our computer system and destroys the stored information is called *virus*. Virus usually corrupts the resources that are connected over the Internet.
- **Bulk e-mail** It is the most common problem of the Internet where the unwanted bulk messages such as subscription mails, advertisement mails, job alert mails etc. are received on the users account. These messages are sometimes frustrating and irritating for the users because the user simply does not require these mails.

14.3.5 The Intranet

Intranet is a private network, which is confined to a single organisation only. This type of computer network allows only the internal users of the organisation to share the resources. However, the users outside the organisation can also access the Intranet but they can do so only if they are authorised. The concept of Intranet was used for sharing the company's information amongst the employees. Certain protocols such as TCP/IP, HTTP, etc. are used by Intranet for enabling the communication between the computer systems. Organisations that use Intranet for transferring the messages generally make use of tunneling, which is a technique used for sending private messages. The website of the Intranet is provided with the firewall, which is a layer that helps in ensuring the security of the information and resources. The only disadvantage of Intranet is that it is relatively insecure as compared to the other networks. The various advantages of using Intranet are as follows:

- It allows the employees of an organisation to access the organisation's information easily and quickly.
- Intranet users can easily communicate with each other within the organisation.
- It is relatively easier to maintain and implement Intranet than the Internet.

14.3.6 Client Server Network (CSN)

CSN basically consists of two computers, client computer and server computer. The client computers are dummy computers, which simply send requests to the server computer, whereas the server computers receive and execute the requests sent by the client computer. CSN is also known as the client server architecture. This architecture is a two-tier architecture, which is divided into two layers. The first layer comprises of the user interface that is located on the client's desktop. On the other hand, the second layer is the database management layer, which is located on the server machine so as to provide services to the clients.

In CSN, a client computer sends a request related to processing of data to the server. The server receives the request from the client computer and processes the data. It then sends the output obtained after the processing of data to the client computer as a response to its request. The following are the advantages of the CSN:

- There is a proper distribution of computing tasks. The task of the client computer is to send request to the server for processing of data. The server receives the request of the client computer, processes the data and sends the result back to the client computer.
- All the important data is stored on the server and only authorised persons are allowed to access the data from the server. As a result, there is greater security of data in a CSN.
- It is easy to manage the data updates as data is stored in a centralised manner on a server.
- Multiple client computers having varying hardware configuration are able to interact with a server.

The following are the disadvantages of the CSN:

- If the server crashes, the complete network breaks down.
- The load on the server increases in a CSN because there are multiple clients sending some request to a server. As a result, the network traffic increases and network congestion occurs.

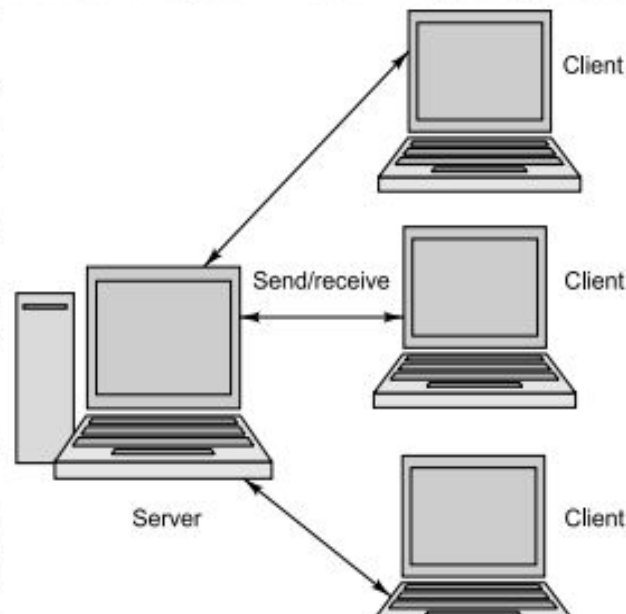


Fig. 14.7 A CSN

14.3.7 Peer-to-peer Network (PPN)

In PPN, there is no separate division as clients and servers. Every computer in the PPN is treated equally and can send as well as receive the messages directly. PPN architecture cannot work under heavy load. This type of architecture is commonly used for file sharing and chatting in real time. Computers connected in this network can easily share their resources with the other computers. PPN is used in a variety of fields such as business, education, telecommunications, military etc. The advantages of PPN are as follows:

- PPN is relatively cheaper than the CSN, as the centralised server is not required.
- PPN is simpler than the CSN because computers connected in the network can communicate efficiently with each other.

The following are the disadvantages of the PPN:

- PPN is less secure because message flows freely between the computers.
- PPN is decentralised as there is no specific space for the storage of files.
- PPN is less expandable as performance and speed of network degrades when more computers are added.

Figure 14.8 shows the PPN.

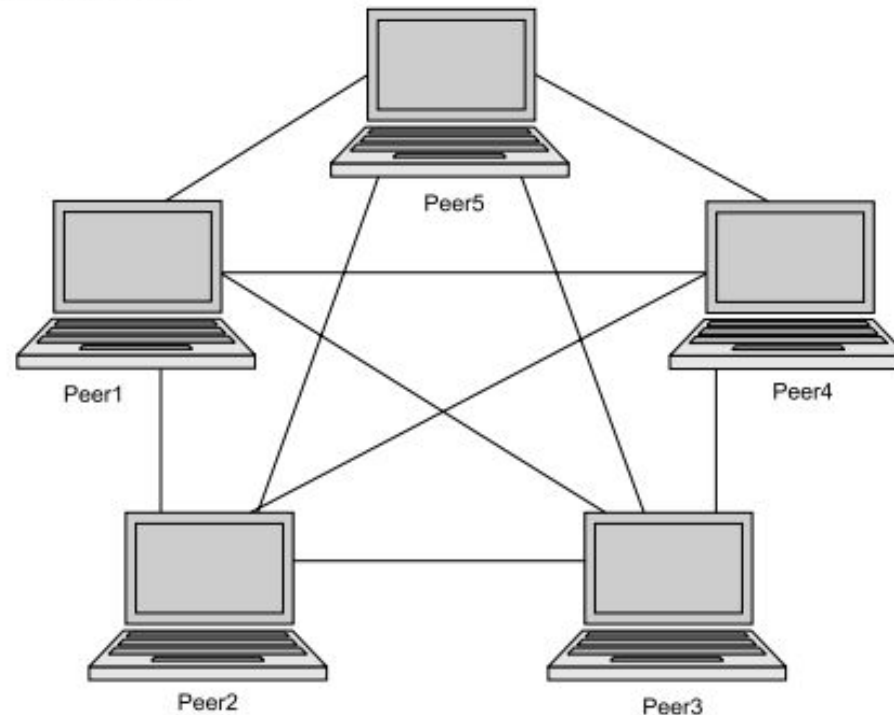


Fig. 14.8 The PPN

14.3.8 Value-added Network (VAN)

VAN is a network that is generally used by the companies privately for Electronic Data Interchange (EDI), which is a standard used for interchanging the business data. In other words, VAN is a network that usually adds the value to the transmitted information by providing value-added services. An example of value added service could be the conversion of data in eXtensible Markup Language (XML) to EDI provided through VAN to specific companies.

Apart from providing value-added services, VAN also provides the auto error detection and correction and protocol conversion services for data modification. Error detection and correction is a technique used for detecting the errors in the transmitted data and then rectifying it. On the other hand, protocol conversion is basically used for converting a message, which has been transferred using a specific protocol to another protocol with the help of protocol converter.

14.4 NETWORK TOPOLOGIES

Network topology refers to the arrangement of computers connected in a network through some physical medium such as cable, optical fibre etc. Topology generally determines the shape of the network and the communication path between the various computers (nodes) of the network. The various types of network topologies are as follows:

- Hierarchical topology
- Bus topology
- Star topology
- Ring topology
- Mesh topology
- Hybrid topology

14.4.1 Hierarchical Topology

The hierarchical topology is also known as tree topology, which is divided into different levels connected with the help of twisted pair, coaxial cable or fibre optics. This type of topology is arranged in the form of a tree structure in which top level contains parent node (root node), which is connected with the child nodes in the second level of hierarchy with the point-to-point link. The second level nodes are connected to the third level nodes, which in turn are connected to the fourth level nodes and so on. Except the top-level node, each level node has a parent node.

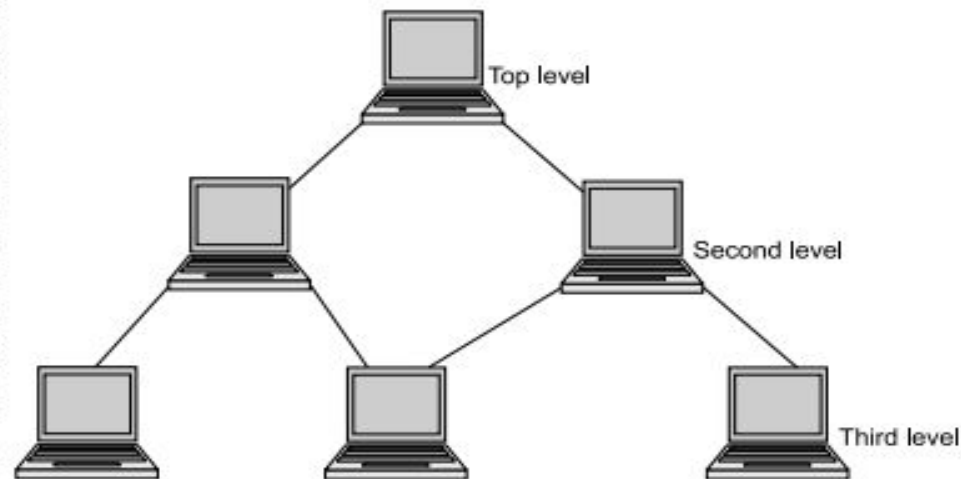


Fig. 14.9 The hierarchical topology

The number of point-to-point links in the hierarchical type of topology is generally one less than the total number of nodes in the structure. The hierarchical topology is symmetrical, having a fixed branching factor, f , associated with each node. The branching factor is the number of point-to-point links between the levels of hierarchy. Figure.14.9 shows the arrangement of computers in hierarchical topology.

Advantages of hierarchical topology are:

- The hierarchical topology is generally supported by most hardware and software.
- In the hierarchical topology, data is received by all the nodes efficiently because of point-to-point link.

The following are the disadvantages of hierarchical topology:

- In the hierarchical topology, when the root node fails, the whole network crashes.
- The hierarchical topology is difficult to configure.

14.4.2 Linear Bus Topology

In the linear bus topology, all the nodes are connected to the single backbone or bus with some medium such as twisted pair, coaxial cable etc. When a node wants to communicate with the other nodes in the network, it simply sends a message to the common bus. All the nodes in the network then receive the message but the node for which it was actually sent only processes it. The other nodes discard the message. Figure 14.10 shows the arrangement of computers in the linear bus topology.

Advantages of linear bus topology are:

- The linear bus topology usually requires less cabling.
- The linear bus topology is relatively simple to configure and install.
- In the linear bus topology, the failure of one computer does not affect the other computers in the network.

The following are the disadvantages of linear bus topology:

- In the linear bus topology, the failure of the backbone cable results in the breakdown of entire network.
- Addition of computers in the linear bus topology results in the performance degradation of the network.
- The bus topology is difficult to reconstruct in case of faults.

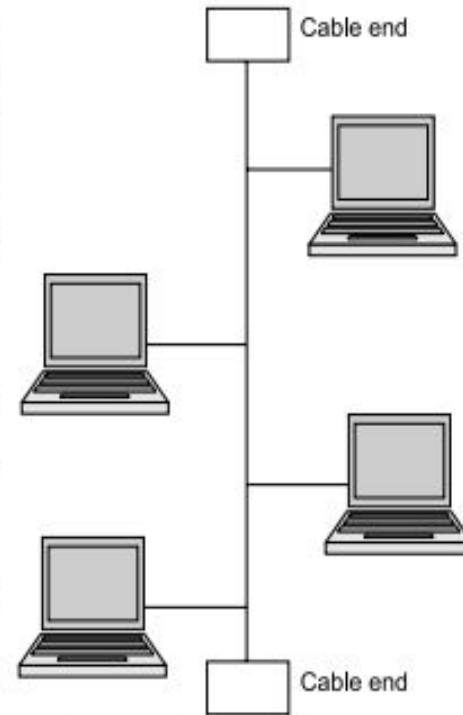


Fig. 14.10 A linear bus topology

14.4.3 Star Topology

In the star topology, all the nodes are connected to a common device known as hub. Nodes are connected with the help of twisted pair, coaxial cable or optical fibre. When a node wants to send a message to the other nodes, it first sends the message to the hub, which in turn forwards the message to the intended node. Each node in the network is connected with a point-to-point link to the centralised hub. The task of hub is to detect the faulty node present in the network. On the other hand, it also manages the overall data transmission in the network. Figure 14.11 shows the arrangement of computers in star topology.

Advantages of star topology are:

- This topology allows easy error detection and correction.
- In the star topology, the failure of one computer does not affect the other computers in the network.
- Star topology is easy to install.

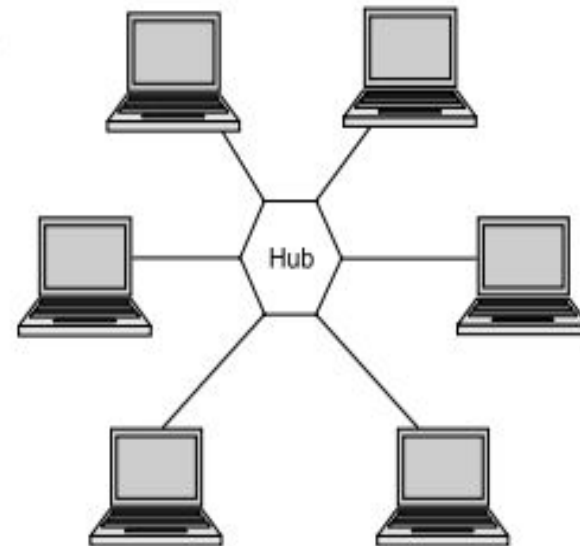


Fig. 14.11 A star topology

The following are the disadvantages of star topology:

- In the star topology, the hub failure leads to the overall network crash.
- The star topology requires more amount of cable for connecting the nodes.
- It is expensive due to the cost of hub.

14.4.4 Ring Topology

In the ring topology, the nodes are connected in the form of a ring with the help of twisted pair. Each node is connected directly to the other two nodes in the network. The node, which wants to send a message, first passes the message to its consecutive node in the network. Data is transmitted in the clockwise direction from one node to another. Figure 14.12 shows the arrangement of computers in the ring topology. Each node incorporates a repeater, which passes the message to next node when the message is intended for another node.

Advantages of ring topology are:

- Each node has an equal access to other nodes in the network.
- Addition of new nodes does not degrade the performance of the network.
- Ring topology is easy to configure and install.

The following are the disadvantages of ring topology:

- It is relatively expensive to construct the ring topology.
- The failure of one node in the ring topology affects the other nodes in the ring.

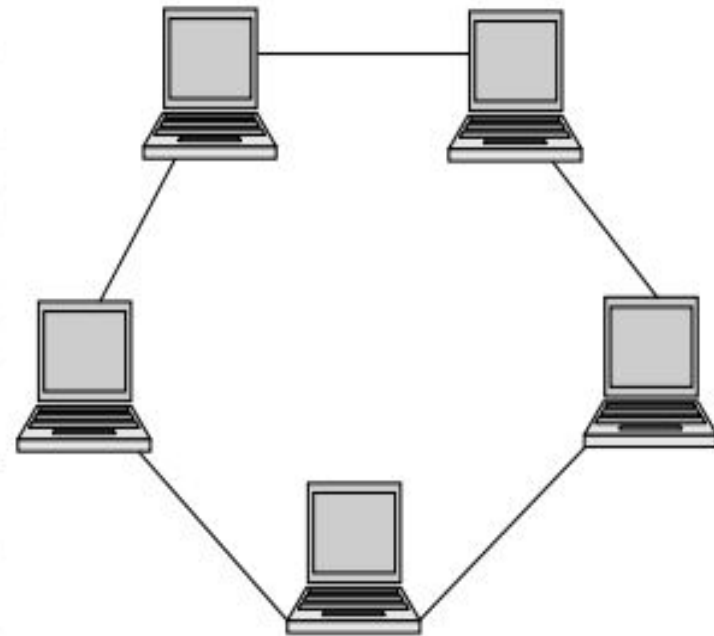


Fig. 14.12 A ring topology

14.4.5 Mesh Topology

In mesh topology, each computer is connected to every other computer in point-to-point mode as shown in Fig. 14.13. For example, if we have four computers, we must have six links. If we have n computers, we must have $n(n-1)/2$ links. A message can take several possible paths to reach a destination.

Advantages of mesh topology are:

- Message delivery is more reliable.
- Network congestion is minimum due to large number of links.

The following are the disadvantages:

- It is very expensive to implement.
- It is very difficult to configure and install.

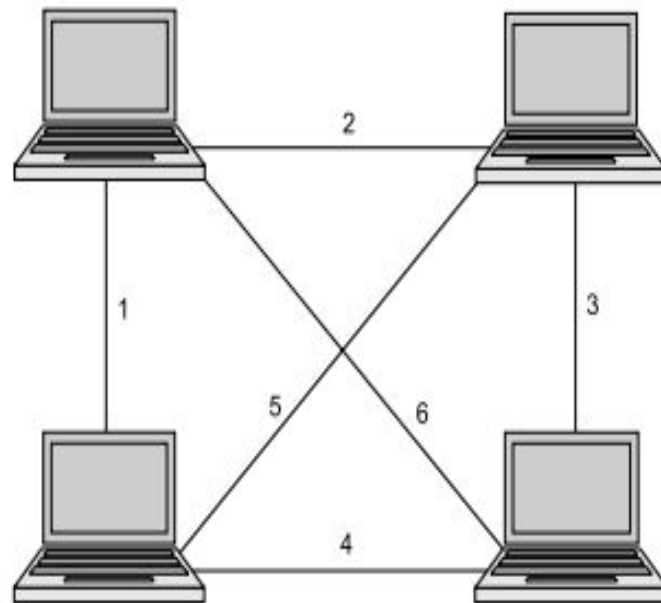


Fig. 14.13 Mesh topology

14.4.6 Hybrid Topology

The hybrid topology is the combination of multiple topologies, used for constructing a single large topology. The hybrid topology is created when two different network topologies are interconnected. If two ring topologies are connected then the resultant topology is not the hybrid topology. On the other hand, if the ring topology is connected to the bus topology then the resulting topology is called the hybrid topology. This topology generally combines the features of the two topologies and is therefore more effective and efficient than the individual topologies. Figure 14.14 shows a typical arrangement of computers in hybrid topology.

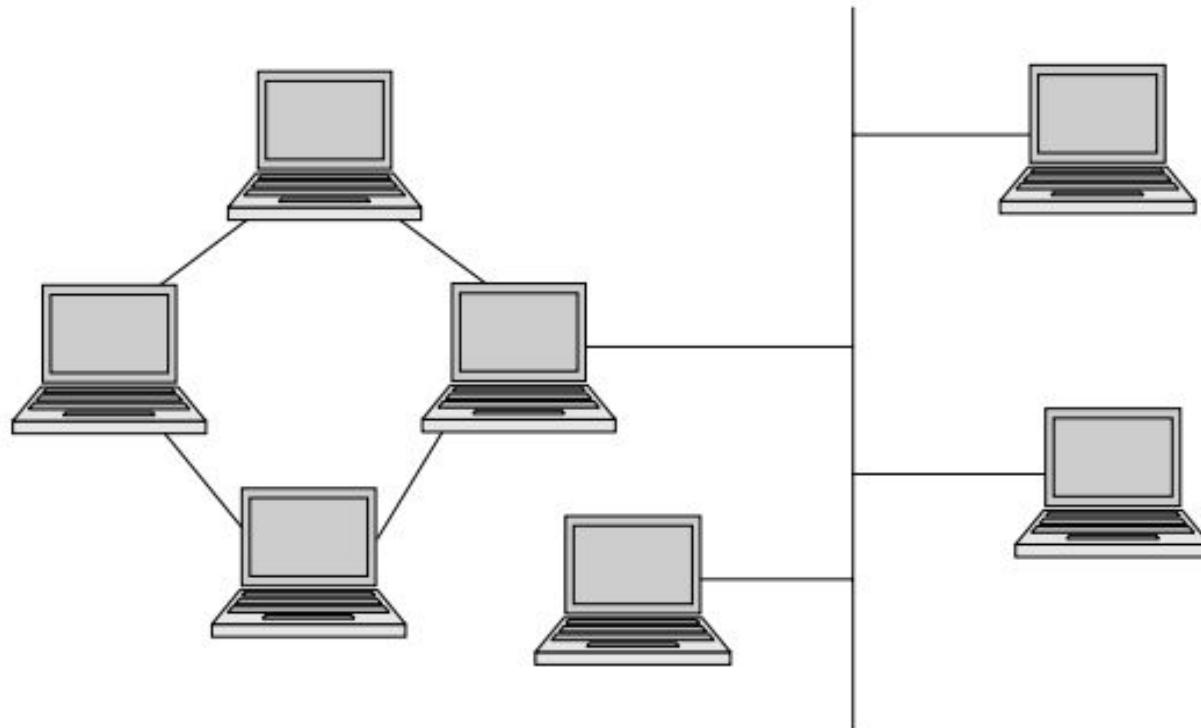


Fig. 14.14 A hybrid topology

Advantages of hybrid topology are:

- The hybrid topology is more effective as it uses multiple topologies.
- The hybrid topology contains the best and efficient features of the combined topologies from which it is constructed.

The following are the disadvantages of hybrid topology:

- The hybrid topology is relatively more complex than the other topologies.
- The hybrid topology is difficult to install and configure.

14.4.7 Network Media

Network media refers to the physical media used to connect the computer nodes together. There are many types of transmission media, the most popular being cables, optical fibre, microwave and satellite.

- **Cables** Cable is one of the easiest methods of transferring messages from one place to another. Some of the commonly used cables are telephone lines, twisted-pair cables and coaxial cables.

Twisted-pair cables are insulated and twisted around each other in order to reduce noise (unwanted signal). They are generally used to connect hubs to nodes in star topology. Coaxial cables have single copper conductor with a shield around it. They are very robust and used for crossing large distances such as the ocean.

- **Optical fibre** An optical fibre is a thin strand of glass that transmits pulsating beams of light rather than electric current. These light pulses carry information. Fibre-optic cables can thus carry information in digital form. They can carry enormous amount of messages at extremely fast speed. These are used for long distance communications. It is a highly secured transmission medium.
- **Microwave** Communication through microwave can be used when the transmitting and receiving ends are located at a large distance from one another. Microwaves are used to transmit analog signals. WANs often use microwave links to connect LANs together. This mode of transmission is greatly affected by the weather conditions.
- **Satellite** Satellites are used for receiving and transmitting analog signals globally. We can send data from one country to another with the help of satellites. WANs that cover long distances often use satellites for linking LANs.

14.5 NETWORK PROTOCOLS AND SOFTWARE

In order to share data between computers, it is essential to have appropriate network protocols and software. With the help of network protocol, computers can easily communicate with each other and can share data, resources etc.

14.5.1 Network Protocol

Network protocols are the set of rules and regulations that are generally used for communication between two networks. Any two networks communicate with each other by sending and receiving messages in the form of packets. The technique that is used separating a message into packets is called packet switching. Each packet contains the address of the computer from which the message was sent and also of the computer, which will receive the message. In order to send the packet, routers and switches are connected over the network path that forwards the packet to the intended receiver. Using network protocol, the following tasks can be performed:

- Identification of the type of the physical connection used
- Error detection and correction of the improper message
- Initiation and termination of the communication session
- Message formatting

Some of the commonly used network protocols are Hyper Text Transfer protocol (HTTP), Simple Mail Transfer Protocol (SMTP), File Transfer Protocol (FTP), Transmission Control Protocol/ Internet Protocol (TCP/IP), Telecommunications Network (Telnet), Domain Name System (DNS) etc.

14.5.2 Network Software

Network software refers to the set of instructions that are given to the computers connected in a network to perform different tasks. These instructions are given in the form of a program, which is usually written in some programming language such as C++, Java, C etc. The network software generally provides support to the computers connected in a network so that they can communicate with each other by exchanging

information. Network operating system software is the type of network software that basically controls the network traffic, access to the network resources such as printers, files, etc. This type of software generally provides multiuser, multitasking facilities so as to ensure effective communication between the computers over the network. It also establishes the communication between multiple computers that are performing a single task.

14.6 APPLICATIONS OF NETWORK

Network is the system of computers generally linked together to enable the flow of data between the interconnected computers. With the help of network, we can access the data remotely. Some of the applications of a network are:

- **Data sharing** It is the capability of sharing data with multiple users over the network. In order to share the data among multiple users, it is generally stored on the servers. Different applications can therefore access the data from these servers easily. A special software locking mechanism is maintained over the network, so as to prevent multiple users from modifying the data. The sharing of data among the multiple users in an interactive way is known as data conferencing.
- **Remote data access** Remote data access is the process of accessing the data from remote location in an efficient way. Different software programs are used to provide an interface to the end users for accessing the data remotely.
- **Resource sharing** Resource sharing is the process of sharing the resources such as storage devices, input/output devices, etc. over the network. For example, printer can be shared among the computers connected in a network by attaching it to the server computer. When the users want their documents to be printed they can simply give the print command from their computers and get their document printed. The sharing of a printer on a network is relatively cheaper than attaching separate printers to the individual computers connected in the network.
- **Personal communication** The far-reaching applications of networks are electronic mail and teleconferencing. These applications allow individuals as well as organisations to use networks for exchanging messages (written, voice and video) extensively. Electronic mail enables a person to send and receive instant messages over a computer network with the help of Internet. On the other hand, teleconferencing enables real-time communication over a distance by allowing people at different locations to communicate with each other by seeing the video picture of people at other sites. This is also referred to as virtual meeting.



Thanks

